2024 Fourth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT) | 979-8-3503-4367-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICAECT60202.2024.10468789

Machine Learning Model to Predict Results of Law Cases

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***Abstract* - The judicial system is an important component of the nation's democratic system, protecting the rights and liberties of its residents while preserving the rule of law. With time, many necessary changes are made in the judiciary system to maintain peace, trust, and order in the country. But court hearings and proceedings take too much time for a decision. Despite the fact that the legal industry is developing faster than ever with the aid of developing technologies, there are still many unexplored areas and there is always potential for improvement. In this paper, we present a simplified approach, "*AI in Law Practice*". The model is developed by utilizing, the most disruptive technology - Machine Learning. The dataset is stored in IPFS for legal and ethical considerations. The algorithm can forecast case results, giving departments and attorneys useful information. Predicting the case output with accuracy is the model's problem. The F1-score, accuracy, precision, recall, support, and recall are used to evaluate the system's performance, which also demonstrates the system's practical applicability. The model is trained on 3304 U.S. Supreme Court cases and achieves 95% accuracy. The model that is currently being built will be utilized by legal practitioners, law departments, end users, etc.**

Keywords - Machine Learning, IPFS, AI, Supreme Court, Appeals, Judgements, Case Prediction, Natural Language Processing.

I. INTRODUCTION

Rule regulations, rights, and duties are very important parts of law. These are maintained by the legal system of the country. The system should be trusted and provide the right justice to the society. But justice takes months, years, or sometimes decades. This is because of the inefficient, complex, or slow legal system of the nation [1]. As of July 20, 2023, there are 5.02 crore cases pending in the nation's courts, including the Supreme Court, 25 high courts, and subordinate courts [2]. When judicial effectiveness is calculated as the number of judges per case filed, a drastically different picture is seen. This analysis suggests that there might be additional causes for India's issue with judicial delays. It can result from a special aspect of Indian law, a shortage of funding for other judicial departments, or some other unidentified factor [1]. A few of the reasons why cases are pending in court are the dearth of judges and judicial officers, the court's support staff and physical infrastructure, the complexity of the facts at hand, the type of evidence, and the cooperation of various parties including the Bar, investigating agencies, witnesses, and litigants [2].

The integration of artificial intelligence (AI) into various domains has transformed industries and professions, promising increased efficiency, precision, and innovation. Within the legal sector, AI's potential to revolutionize the practice of law is becoming increasingly evident. One of the most promising applications of AI in law practice is the development of AI-based Case Prediction Systems. These algorithms are ready to change the way legal practitioners approach their work since they can forecast a variety of legal case outcomes, from verdicts to settlement probability. In a world where legal systems grapple with increasing caseloads, resource limitations, and the complexities of the modern legal landscape, the introduction of AI-driven predictive analytics holds the promise of optimizing decision-making processes, improving resource allocation, and enhancing the accessibility and fairness of legal services.

As legal practitioners and stakeholders seek innovative solutions to the challenges facing the legal profession, AI-based Case Prediction Systems stand out as a transformative tool with the potential to revolutionize the way law is practiced. Though computer and legal scientists have recently investigated a range of possibilities and opportunities to apply Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP) techniques and technologies into the legal system, the legal system has remained largely isolated from the Information Technology (IT) world [3].

A few forward-thinking scientists have embarked on the difficult task of applying ML and NLP approaches to construct prediction models that reliably predict court outcomes in an effort to enhance, improve, and elevate the civil justice system. One important objective is to cut down on the amount of time needed for court cases and judgment delivery. In this light, it becomes vital and crucial to construct accurate and efficient legal predictive models in order to better serve society [3]. Prediction models are an additional tool that judges and justices in courts can employ to speed up their decision-making. The prediction models can be used by the attorneys to anticipate the outcome of cases, which will enable them to better plan their defense during case argument.

This research paper begins a thorough investigation of the role and impact of AI in law practice, with a specific focus on Case Prediction Systems. It delves into the significance of AI in predicting legal case outcomes, the underlying technologies that drive these systems, and the opportunities and challenges they present to legal professionals, courts, clients, and the broader legal ecosystem.

Significance of AI in Law Practice

The legal profession, characterized by its reliance on precedent, legal analysis, and intricate argumentation, presents an ideal domain for the application of AI technologies. The sheer volume of legal information, past case records, statutes, and regulations is overwhelming, making it difficult for legal professionals to efficiently examine and predict case outcomes. AI systems, powered by machine learning algorithms and natural language processing (NLP), have the potential to sift through vast datasets, recognize patterns, and generate predictions with a level of speed and precision that surpasses human capabilities [3].

The significance of AI in law practice extends beyond the realm of prediction accuracy. It encompasses the capacity to optimize resource allocation, streamline legal research, enhance decision-making, and ultimately democratize access to legal services. By providing legal professionals with data-driven insights, AI systems can help prioritize cases, allocate resources effectively, and mitigate legal risks, all while reducing costs for clients. This transformative potential not only enhances the practice of law but also addresses longstanding challenges in the legal profession, including backlog reduction, judicial efficiency, and equitable access to justice.

However, given the complexity of the legal system, anticipating court decisions and case outcomes with reasonable explanation with increased accuracy is a challenging and time-consuming task (Ruhl et al., 2017). This calls for intensive use of general IT, NLP, and ML for data extraction, cleansing, modeling, and engineering, as well as for model development, training, and testing [4].

As a result, we also strive to contribute to this difficult work by creating an effective prediction model that can accurately forecast rulings, appeals, and verdicts from the US Supreme Court. Additionally, we describe in this work our created machine learning-enabled legal prediction model that accurately forecasts decisions made by the US Supreme Court. The model has a 95% accuracy rate after being trained and evaluated on 3304 cases from the Supreme Court.

A. MACHINE LEARNING

Machine learning is an emerging field of artificial intelligence that provides the capability to machine, to learn from experience, think and act like human without any human intervention. It is a popular technique for data analysis that is used to find patterns or forecasts in large amounts of data gathered from various circumstances [5]. Prior to using machine learning in practice, data collection, appropriate algorithm selection, model training, and model validation must be completed. Choosing the appropriate algorithm is crucial for these processes. Supervised and unsupervised learning are the two main types of machine learning technology [6]. The main difference between these two kinds is whether or not labels are present in the datasets. In supervised learning, prediction functions are inferred from the labeled training datasets. Every training instance contains input values as well as expected output values.

Supervised learning algorithms search for connections between the input and output values in order to build a prediction model that will foresee the result based on the relevant input data. Many algorithms have been developed for supervised learning, such as naive Bayes, decision tree (DT), support vector machine (SVM), random forest (RF), and k-nearest neighbor (KNN), and linear regression. Supervised learning can be used for both data categorization and regression.

Conversely, unsupervised learning is commonly employed to handle unlabeled data and use unlabeled training datasets to address a range of pattern identification problems. Dimensionality reduction and clustering are the main techniques used in unsupervised learning to divide training data into discrete groups according to their distinctive characteristics [7].

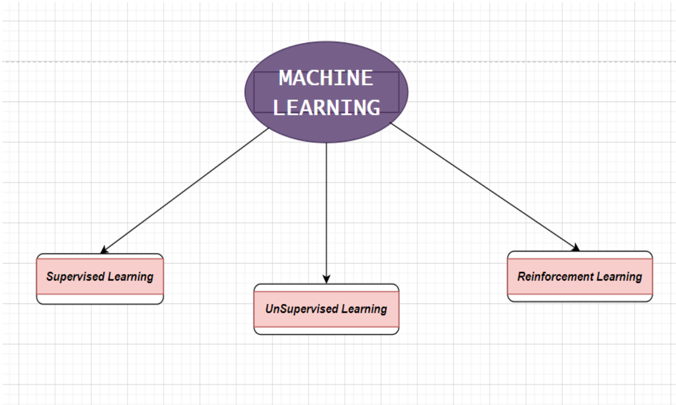

Fig 1. Machine Learning Types

Nevertheless, it's unclear how many categories there are and what each one means. Consequently, unsupervised learning is often used in association mining and classification [6]. Principal component analysis (PCA), K-means, and other unsupervised machine learning techniques are often used. Furthermore, another category of machine learning methods is called reinforcement learning, which describes a machine's capacity to generalize and provide accurate answers to issues that it hasn't been taught.

B. IPFS

Building upon the fundamentals of P2P networking and content addressing, the Interplanetary File System (IPFS) is a decentralized cloud storage is the goal of a novel decentralized storage system. [8]. IPFS is an intriguing large-scale operational network to analyze, with over 230k peers using it each week and serving daily tens of millions of requests. Although it is a fundamental component of many initiatives and studies, little is known about its inner workings, characteristics, or consequences from research.

The main goal of IPFS is to establish a decentralized, worldwide network for file sharing and storage. By enabling files to be stored in various locations, making them resistant to censorship, and guaranteeing availability even if some nodes go offline, it seeks to increase the efficiency and robustness of conventional web protocols. Since its initial release in 2014, the IPFS P2P file-sharing network has grown in popularity as a result of its distinctive characteristics, which provide an alternative to the conventional client-server design. By improving the scalability of DApps, the protocol has already completely changed the cryptocurrency space and garnered popularity across a number of sectors, including media and banking.

IPFS has the ability to fundamentally change how we share, save, and access information online. As the technology develops further, we may anticipate even more breakthroughs and uptake in the future.

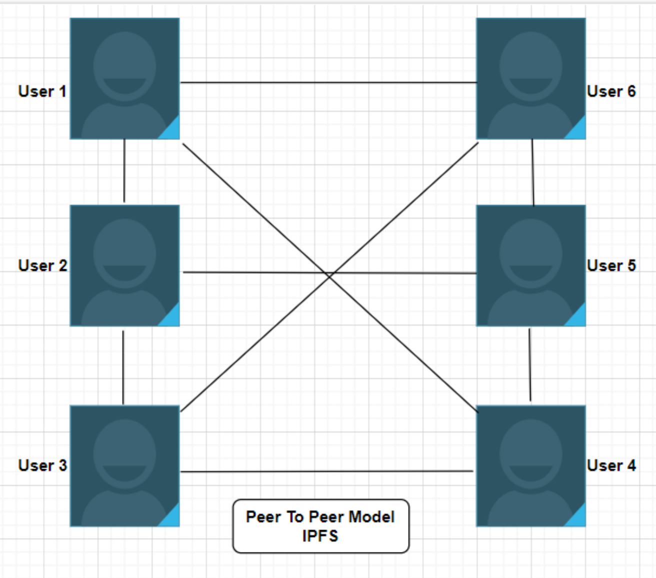

Fig 2. IPFS

This is how the rest of the paper is organized. The relevant literature is covered in Section 2 with a focus on court decision forecasts. The problem statement in Section 3 motivates us to work on this study. The suggested work is discussed in Section 4. Section 5 provides an explanation of the outcome. The fascinating topic of using ML and NLP to the legal system is expanded upon in Section 6. In section 7, the study concludes with a quick summary of the next steps.

II. LITERATURE SURVEY

The paper introduces a The Indian Supreme Court's rulings are accurately predicted by the recently created machine learning (ML)-enabled legal prediction model and its working prototype, eLegPredict [3]. With training and testing on 3072 Supreme Court cases, eLegPredict has attained an accuracy of 76% (F1-score). The authors contend that the legal community would greatly benefit from and be interested in legal predictive models. These models can be used by stakeholders, particularly judges and lawyers, to forecast case outcomes and enhance their future activities, such as expediting the decision-making process, bolstering the defense, and providing support for arguments.

But correctly anticipating court rulings and case results is a difficult process that requires a number of intricate stages, including locating pertinent bulk case papers, extracting, cleaning, and engineering data. eLegPredict addresses these challenges by using a novel ML approach that incorporates both legal and non-legal features of a case [3]. The legal features include the type of case, the parties involved, the relevant statutes and case laws, and the arguments of the lawyers. The non-legal features include the composition of the bench, the history of the judges, and the public opinion on the case.

eLegPredict is a feature that helps users by automatically reading through documents with fresh case descriptions and producing a forecast as soon as they are deposited into a specified directory. The authors come to the conclusion that eLegPredict is a technology that holds great promise for enhancing the efficacy and efficiency of the Indian legal system.

Katz, D. M., Bommarito, M. J., & Blackman, J. (2017). "A General Approach for Predicting the Behavior of the Supreme Court of the United States." [9] presents an innovative method for forecasting the actions of the US Supreme Court. Leveraging data-driven techniques, the study examines a vast dataset of historical Supreme Court decisions, legal precedents, and contextual factors to develop a predictive model.

The model is designed to forecast the Court's decisions, including voting outcomes and opinions authored by justices, with an astounding level of precision. Natural language processing (NLP) and machine learning techniques are used by the authors to show how this method can change how people perceive the Court's decision-making process. The research offers valuable insights into the integration of AI and data analytics in legal studies and the capacity to anticipate judicial decisions within the highest court in the nation, with profound implications for legal practice and scholarship.

In order to predict the verdicts from Brazilian courts, André Lage-Freitas et al. (2019) [10] conduct research, build a prediction technique, code, and produce an operational prototype that, when tested and trained on 4,043 Brazilian court cases, exhibits a 79% accuracy rate.

In 2019, O'Sullivan and Beel [11] produce machine learning (ML)-enabled models to forecast whether rulings under the ECHR breach any human rights article. The authors employ word (echr2vec) and paragraph (doc2vec) embeddings to improve performance, and the result is an overall test accuracy of 68.83%.

Pillai and Chandran (2020) [12] investigate the use of CNN and NLP, with a focus on Bag of Words, to extract key terms from Indian court rulings, categorize them as bailable or non-bailable, and then forecast which category they will fall into. Their prediction model has an accuracy of about 85%.

In order to determine if the rulings made by the Turkish Constitutional Court on specific instances breach any public morals or freedom of expression, Sert et al. [13] (2021) use artificial intelligence (AI) techniques combined with natural language processing (NLP) word embedding. The authors claim that their prediction accuracy is about 90%.

The paper "Automatic Judgment Prediction via Legal Reading Comprehension" by Shangbang Long, Cunchao Tu, Zhiyuan Liu, and Maosong Sun [14] suggests an innovative method of employing legal reading comprehension (LRC) for automatic decision prediction. Automatic judgment prediction is the task of predicting the outcome of a legal case based on the case materials, such as the fact description, plaintiffs' pleas, and law articles. The majority of current techniques use a framework for text classification, which is unable to capture the intricate relationships between the various kinds of case materials.

LRC formalizes the task of predictive automatic judgment as a reading comprehension exercise. In LRC, the model is given the case materials as input and asked to answer a question about the outcome of the case. The question is typically phrased as a multiple-choice question, such as "What is the most likely outcome of the case?"

The authors present a brand-new LRC model dubbed AutoJudge that effectively captures the intricate semantic relationships between the laws, pleas, and facts. On a real-world civil case dataset, AutoJudge outperforms state-of-the-art models by a significant margin [14]. The authors argue that LRC is a promising approach to automatic judgment prediction because it allows the model to reason about the case materials in a more comprehensive and informative way than traditional text classification methods.

Aastha Budhiraja and Kamlesh Sharma's study "Correlation of Language Processing and Learning Techniques for Legal Support System" [15] explores the application of machine learning to the prediction of case outcomes and court rulings. The authors point out that while this is a laborious and intricate procedure, language processing and learning techniques can help to increase its accuracy and efficiency.

The authors provide a classifier that combines the Att-BLSTM model with Text CNN. This approach can automatically extract richer features from legal documents by emphasizing specific traits and fully accounting for full-text information. The suggested model performed better in trials than SVM-TFIDF, deep pyramid convolution neural network (DPCNN), and hierarchical attention network (HAN) document categorization models.

The authors conclude that their proposed model is a method that shows promise for forecasting court rulings and case results. They argue that this could be a valuable tool for legal professionals, helping them to make better decisions and improve the efficiency of the legal system. The paper is relevant to the field of legal support systems, as it provides a new approach to predicting legal judgments and case outcomes [15]. This could be used to develop more accurate and efficient legal support systems, which could benefit both legal professionals and the public.

Table 1. Literature Survey

Title	Author	Year
Predicting Indian Supreme Court Judgments, Decisions, or Appeals: eLegalls Court Decision Predictor (eLegPredict)	Sharma, S. K., Shandilya, R., & Sharma, S	2023
Harnessing legal complexity	Ruhl, J. B., Katz, D. M., & Bommarito, M. J.	2017
A general approach for predicting the behavior of the Supreme Court of the United States	Katz, D. M., Bommarito, M. J., & Blackman, J.	2017
Predicting Brazilian court decisions	Lage-Freitas, A., Allende-Cid, H., Santana, O., & de Oliveira-Lage, L.	2019
Predicting the outcome of judicial decisions made by the European court of human rights.	O'Sullivan, C., & Beel, J.	2019
Verdict Prediction for Indian Courts Using Bag of Words and Convolutional Neural Network.	Pillai, V. G., & Chandran, L. R.	2020
Using Artificial Intelligence to Predict Decisions of the Turkish Constitutional Court	Sert, M. F., Yıldırım, E., & Haşlak	2021
Automatic judgment prediction via legal reading comprehension.	Long, S., Tu, C., Liu, Z., & Sun, M.	2019
Correlation of Language Processing and Learning Techniques for Legal Support System	A. Budhiraja and K. Sharma	2022

III. PROBLEM STATEMENT

Building an AI model for precise and automatic case prediction and legal research is the issue this paper attempts to solve.

Key Problems

1. Prediction Accuracy
2. Data Collection and Management
3. Feature Engineering
4. Machine Learning Models
5. Ethical and Legal Considerations

Motivation for the proposed work

- **Efficiency Enhancement:** The legal system often deals with a large number of cases, leading to inefficiencies in resource allocation and time management.
- **Risk Mitigation:** For clients, understanding the potential outcomes of their legal cases in advance can help them make informed decisions.
- **Judicial Efficiency:** Courts and judges can benefit from AI-based case prediction by managing their caseloads more efficiently. It can help prioritize cases, reduce backlog, and streamline court proceedings.
- **Consistency in Decision-Making:** AI can provide consistent and objective predictions, reducing potential for human bias or subjectivity in legal decision-making.
- **Enhanced User Experience:** Developing an intuitive interface improves user interaction and accessibility for diverse stakeholders.
- **Enhanced Access to Justice:** By optimizing case management, AI can contribute to the overall accessibility and affordability of legal services, potentially expanding access to justice for underserved populations.

IV. PROPOSED WORK

Machine Learning is the basis of the suggested model. It also uses Natural Language Processing and its main aim is to predict the case outcome that which party should be the winner on the basis of facts and decisions. In order to prepare this model, we used Python packages like scikit-learn, matplotlib, NumPy etc. The dataset is stored in decentralized storage system i.e., IPFS to maintain the confidentiality of the data. The legal data is crucial and it is important to handle it carefully without any legal and ethical issues. The models used here are Logistic Regression, KNN, Random Forest and XGB on the US Supreme Court Dataset.

Figure 3 shows the complete flow of ML model. The figure clearly demonstrates the main parts of the model that contribute to this paper. The content of current Supreme Court rulings has to be cleaned up because it contains a significant amount of language that doesn't help with prediction formation. The PDF copies of the court rulings are read as input and cleaned using natural language processing (NLP), as can be seen in architecture. The most effective vector data is then determined by performing additional necessary NLP operations, which are then included into machine learning (ML). This section goes into further detail about the intermediate NLP processes.

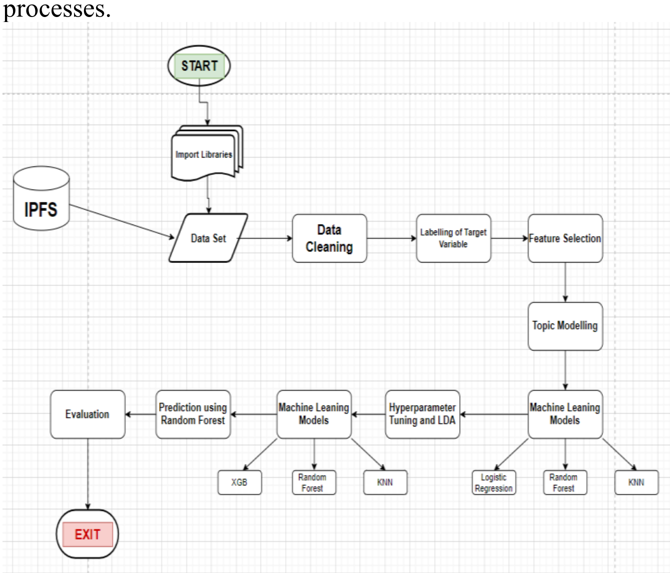

Fig 3. Flow Chart

MODEL DESCRIPTION AND PROCEDURE

IPFS: A distributed file system protocol called the InterPlanetary File System, or IPFS for short, proposes to create a decentralized method for transferring and storing hypermedia. This peer-to-peer hypermedia protocol seeks to improve the efficiency and resilience of online information sharing and access. We have stored the dataset in IPFS for ethical and legal considerations. It stored the data and generate a hash code called as CID. It provide high latency and permanence of data.

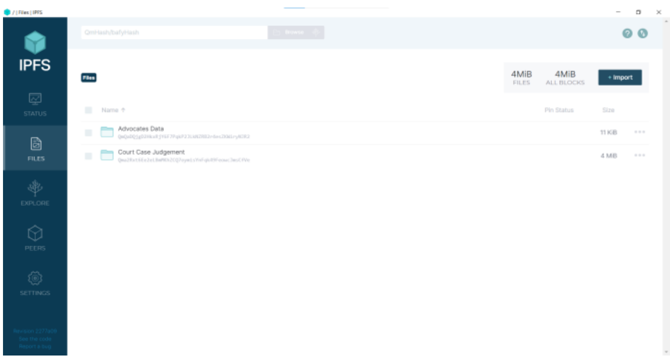

Fig 4. Data stored in IPFS

Python libraries: Import python libraries such as sklearn, matplotlib, NumPy, pandas, NLTK etc., for data cleaning, data manipulation, data visualization, NLP operations, model training and testing.

Dataset: We have used US Supreme Court dataset that contain 3304 cases till 2021. The target variable is first_party_winner that we need to predict, it will win or not.

	Unnamed: 0	ID	name	href	docket	term	first_party	second_party	facts	facts_len	majority_vote	minority_vote	first_party_winner	decision_type	disposition	issue_area
1	1	50613	Stanley v. Illinois	https://api.oyez.org/cases/1971/70-5014	70-5014	1971	Peter Stanley, Sr	Illinois	<p>Joan Stanley had three children with Peter ...	757	5	2	True	majority opinion	reversed/remanded	Civil Rights
2	2	50623	Giglio v. United States	https://api.oyez.org/cases/1971/70-29	70-29	1971	John Giglio	United States	<p>John Giglio was convicted of passing forged...	495	7	0	True	majority opinion	reversed/remanded	Due Process
3	3	50632	Reed v. Reed	https://api.oyez.org/cases/1971/70-4	70-4	1971	Sally Reed	Cecil Reed	<p>The Idaho Probate Code specified that "male...	378	7	0	True	majority opinion	reversed/remanded	Civil Rights
4	4	50643	Miller v. California	https://api.oyez.org/cases/1971/70-73	70-73	1971	Marvin Miller	California	<p>Miller, after conducting a mass mailing cam...	305	5	4	True	majority opinion	vacated/remanded	First Amendment
5	5	50644	Kleindienst v. Mandel	https://api.oyez.org/cases/1971/71-16	71-16	1971	Richard G. Kleindienst, Attorney General of th...	Ernest E. Mandel, et al.	<p>Ernest E. Mandel was a Belgian professional...	2282	6	3	True	majority opinion	reversed	First Amendment

Fig 5. Data set

NLP modelling and data preparation is one of the main responsibilities of this activity. It aids in the reading of all 3304 case documents (in PDF file format), runs a number of intricate intermediary subroutines and processes, and finally generates clean vector data with the proper labelling. The following is a summary of key natural language processing (NLP) techniques that were employed in this work to increase productivity and obtain better semantic representation of words: word stemming, digit removal, stop word removal, removal of smaller words (less than three characters), and conversion of all words to lower case, which enhances word recognition. Taking everything into account, the information that is unnecessary for the decision-making process is removed.

After tokenizing the file, n-grams—where n is a number between 1 and 4—are generated. In order to capture the relevance of each word in the document and corpus as a whole, the textual representation of the words must be quantified into a numerical form for modelling purposes. To this end, the effective TF-IDF (Term Frequency-inverse Document Frequency) technique is employed. The words with lesser frequency (<10%) are ignored by the TF-IDF due to its tuning. By doing this, data can be further refined and only highly representative data is retained. We also create a mechanism as part of the NLP modelling process that scans the target sentences of the case document, automatically capturing the decision categorization of each document in the corpus and appending it as a label to its tokenized vector representation. It also generates a clean CSV data file at the end.

ML models: The following top classifiers are employed in this work: Random Forest, KNN, eXtreme Gradient Boosting (X Gradient Boost), Logistic Regression, and KNN. Our prediction model is built on supervised machine learning classifiers.

Readers are referred to other online resources for the extensive scientific, mathematical, and technical considerations of these classifiers that are outside the purview of this study. Every classifier has its hyperparameters meticulously adjusted for optimal performance. To train and test the model, the whole dataset is split into 80% and 20% ratios, respectively. Python, the Scikit-learn ML package, and the Natural Language Toolkit (NLTK) for NLP are the primary technological tools utilized in this work. Upon effectively training the aforementioned classifiers, we discovered that Random

Forest provides good accuracy, at 80%. For prediction, we thus go with this model.
Evaluation: By using Random Forest for prediction, we get the accuracy of 95%. It accurately predicts which party will win the case.

V. Results & Discussion

The dataset used for training and testing our suggested model comprises 3304 cases from the US Supreme Court between 1955 and 2021. Every case contains its unique IDs, the case facts, and the decision result.

The case facts were infrequently included in other comparable datasets, which may be useful for natural language processing applications. Determining a case's conclusion based just on its facts is one possible use case for this dataset.

Target Variable: First Party Winner: If this statement is accurate, it indicates that the first party prevailed; if not, the second party did. Utilizing NLP approaches, construct characteristics from the facts column.

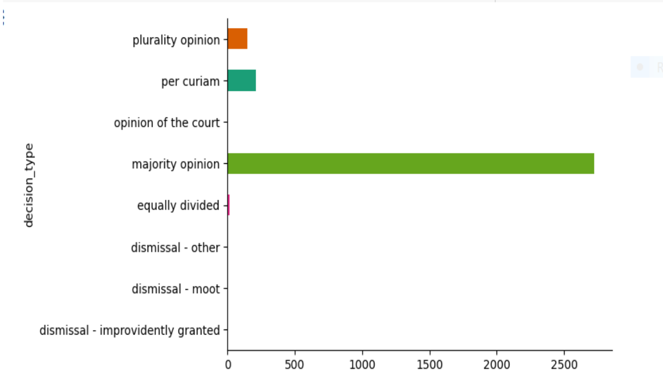

Fig 6. Decision type

Decision type graph shows that on what basis the decision is taken. Most of the decisions are taken on majority opinion of the court.

	facts	facts_clean
0	Joan Stanley had three children with Peter Sta...	joan stanley three child peter stanley stanley...
1	John Giglio was convicted of passing forged mo...	john giglio convicted passing forged money ord...
2	The Idaho Probate Code specified that "males m...	idaho probate code specified male must preferr...
3	Miller, after conducting a mass mailing campai...	miller conducting mass mailing campaign advert...
4	Ernest E. Mandel was a Belgian professional jo...	ernest e mandel belgian professional journalis...
...	...	...
3093	For over a century after the Alaska Purchase i...	century alaska purchase 1867 federal governmen...
3094	Refugio Palomar-Santiago, a Mexican national, ...	refugio palomarsantiago mexican national grant...
3095	Tarahrick Terry pleaded guilty to one count of...	tarahrick terry pleaded guilty one count posse...
3096	Joshua James Cooley was parked in his pickup t...	joshua james cooley parked pickup truck side r...
3097	The Natural Gas Act (NGA), 15 U.S.C. §§ 717–71...	natural gas act nga 15 usc 717717z permit priv...

3098 rows × 2 columns

Fig 7. Clean facts

Facts provided for decision are not cleaned and machine does not understand natural language so we clean the facts using NLP techniques such as tokenization, lemmatization, stop words removal etc.

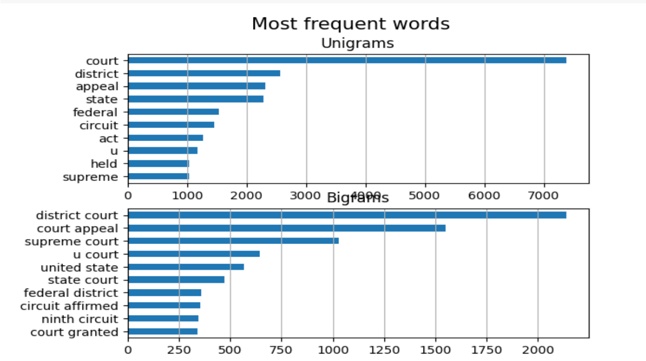

Fig 8. Most frequent words

Most frequent words are the words used in facts. The corpus is difficult to understand for the machine so we have done feature engineering to train our model on each and every word.

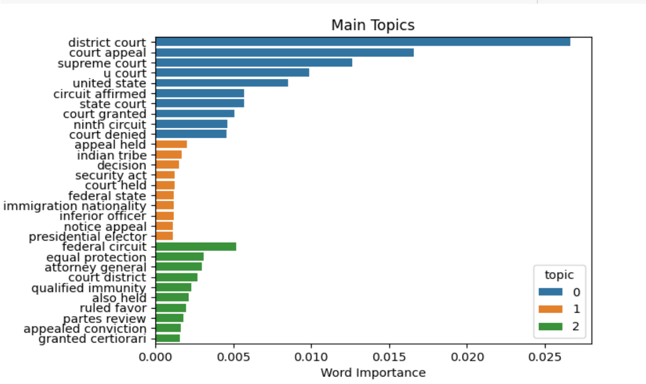

Fig 9. Topic Modelling

Topic modelling is the process in NLP to find out the main topics in the corpus so that our model can understand and learn those topics for further predictions.

Following feature engineering, we use various machine learning algorithms to train our model:

- logistic regression (54%)
- Random Forest (64%)
- KNN (58%)

After performing hyperparameter tuning and LDA, we applied different machine learning models:

- KNN (63%)
- Random Forest (80%)
- XG Boost (66%)

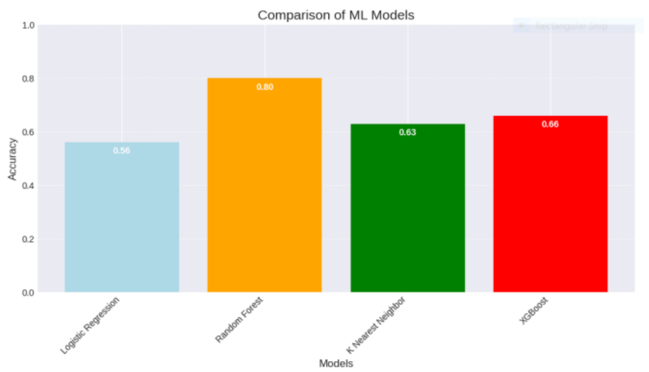

Fig 10. Comparison of ML Models

Random Forest gives good accuracy so we further move on with it and predict the case outcomes using user input.

Table 2. Classification Report

	precision	recall	f1-score	support
0	0.97	0.92	0.95	252
1	0.95	0.98	0.97	406
accuracy			0.96	658
macro avg	0.96	0.95	0.96	658
weighted avg	0.96	0.96	0.96	658

Accuracy: 0.958966565349544

Table 2 shows the evaluation parameter with accuracy of 95%. This demonstrates how well the suggested model predicts the case's conclusion.

```
Case 1: The predicted winner is Second Party
Case 2: The predicted winner is First Party
Case 3: The predicted winner is First Party
Case 4: The predicted winner is Second Party
Case 5: The predicted winner is First Party
Case 6: The predicted winner is Second Party
Case 7: The predicted winner is First Party
Case 8: The predicted winner is First Party
Case 9: The predicted winner is First Party
Case 10: The predicted winner is Second Party
Case 11: The predicted winner is First Party
Case 12: The predicted winner is Second Party
Case 13: The predicted winner is Second Party
Case 14: The predicted winner is Second Party
Case 15: The predicted winner is First Party
Case 16: The predicted winner is First Party
Case 17: The predicted winner is Second Party
Case 18: The predicted winner is First Party
Case 19: The predicted winner is Second Party
Case 20: The predicted winner is First Party
Case 21: The predicted winner is First Party
Case 22: The predicted winner is First Party
Case 23: The predicted winner is First Party
Case 24: The predicted winner is First Party
Case 25: The predicted winner is Second Party
Case 26: The predicted winner is First Party
Case 27: The predicted winner is First Party
Case 28: The predicted winner is Second Party
Case 29: The predicted winner is First Party
Case 30: The predicted winner is Second Party
Case 31: The predicted winner is First Party
Case 32: The predicted winner is Second Party
Case 33: The predicted winner is First Party
Case 34: The predicted winner is First Party
Case 35: The predicted winner is Second Party
Case 36: The predicted winner is First Party
Case 37: The predicted winner is First Party
Case 38: The predicted winner is Second Party
Case 39: The predicted winner is First Party
Case 40: The predicted winner is Second Party
Case 41: The predicted winner is First Party
```

Fig 11. Final Output

The above figure shows the final output.

VI. Conclusion & Future Scope

Promising developments in the application of artificial intelligence (AI) to case prediction in legal practice have occurred in recent years. By utilizing natural language processing and machine learning methods, AI systems have the ability to analyze vast volumes of legal data, including statutes, case law, and precedents, in order to predict case outcomes and offer valuable insights to legal professionals.

The integration of AI in case prediction offers several advantages, including:

- Enhanced Accuracy: AI models can offer predictions based on historical data and patterns, providing insights into possible case outcomes. This aids attorneys in making more informed decisions.

- Efficiency: AI systems can quickly process and analyze large volumes of legal data, saving time for lawyers and legal professionals in researching and preparing cases.
- Risk Assessment: AI systems can assist in evaluating the potential risks associated with legal strategies, thereby aiding lawyers in developing more effective and successful case strategies.
- Cost-Effectiveness: By streamlining the legal research process, AI can potentially reduce costs associated with prolonged legal procedures and research.

The future scope of AI in Law Practice for case prediction is promising and likely to expand further in the coming years. Several areas indicate the potential growth and advancements:

1. **Refinement of AI Models:** Continued advancements in machine learning and natural language processing will lead to more sophisticated AI models capable of better understanding legal nuances and complexities. This will improve the accuracy of case predictions.
2. **Ethical AI Development:** There will be a strong focus on addressing bias and ethical concerns in AI systems. Efforts will be made to create more transparent and fair AI models to ensure unbiased predictions and decisions.
3. **Customization and Specialization:** AI systems will become more specialized, catering to specific legal domains or case types. This specialization will enable more accurate and tailored predictions for different branches of law.
4. **Increased Integration in Legal Firms:** Law firms will increasingly integrate AI tools into their practice for case prediction, legal research, contract analysis, and other tasks to enhance efficiency and productivity.
5. **Collaboration between AI and Legal Professionals:** The future will likely witness a harmonious collaboration between AI technologies and legal professionals. AI will serve as a tool to support lawyers rather than replace them, aiding in more efficient and informed decision-making.

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